University of Essex
MSc Artificial Intelligence
Module: IA_PCOM7E January 2026
Unit 10: Assessment – Research Summary

Assignment Details

The assignment was to research an application of deep learning that could have an impact on society, whether positive or negative, including an overview of the technology, a synopsis of how it works, and any relevant societal impacts it may have in the future. Please note that the total word count was not specified in the rubric and that there was not an option provided to post the assignment to the module discussion boards.

Research Summary

For background context, my undergraduate major, and first Master's program, focused on archaeology, a field in which it is well-known that potentially thousands of significant prehistoric and historic sites are simply too difficult to discover using the current state of the art for surveying and rating the research value of potential excavation sites (Argyrou and Agapiou, 2022). My view is that deep learning could be used to process satellite imagery, LiDAR scans, aerial photographs, and similar material to seek out signs of promising locations for exploratory excavation projects (Satvik and Mehta, 2024).

In short, the process would involve training convolutional neural networks on labelled examples of known archaeological features, such as building foundations, field boundaries, and similar topographic anomalies never found on their own in the natural world (Satvik and Mehta, 2024). That training would focus on details that are not readily apparent to the human eye or to conventional automated processing methods (Satvik and Mehta, 2024).

This approach would offer the benefits of more thorough, accurate, and rapid detection of preliminary excavation sites at a lower cost than conventional methods that require human surveyors, which, in turn, would allow the most valuable sites to be preserved and studied before they run the risk of being damaged or destroyed by modern construction or industrial activities (Argyrou and Agapiou, 2022; Satvik and Mehta, 2024).

References

Argyrou, A. and Agapiou, A. (2022). A Review of Artificial Intelligence and Remote

Sensing for Archaeological Research. *Remote Sensing*, 14(23), p.6000. doi:https://doi.org/10.3390/rs14236000.

Satvik, V. and Mehta, S. (2024). Revolutionizing Archaeological Discoveries: The Role of Artificial Intelligence and Machine Learning in Site Analysis. *2022 13th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, [online] pp.1–5. doi:https://doi.org/10.1109/icccnt61001.2024.10724869.